

Codemix Skill

An intent-validation safety gate for AI voice agents



ALLOW



CONFIRM



BLOCK

*“Order cancel **mat** karo, bas address change karna hai.”*

We sit between what the voice agent hears and what it is allowed to do.

01

Initiative Overview

INITIATIVE

A safety gate between what an AI voice agent hears and what it is allowed to do, for code-mixed Indian callers.

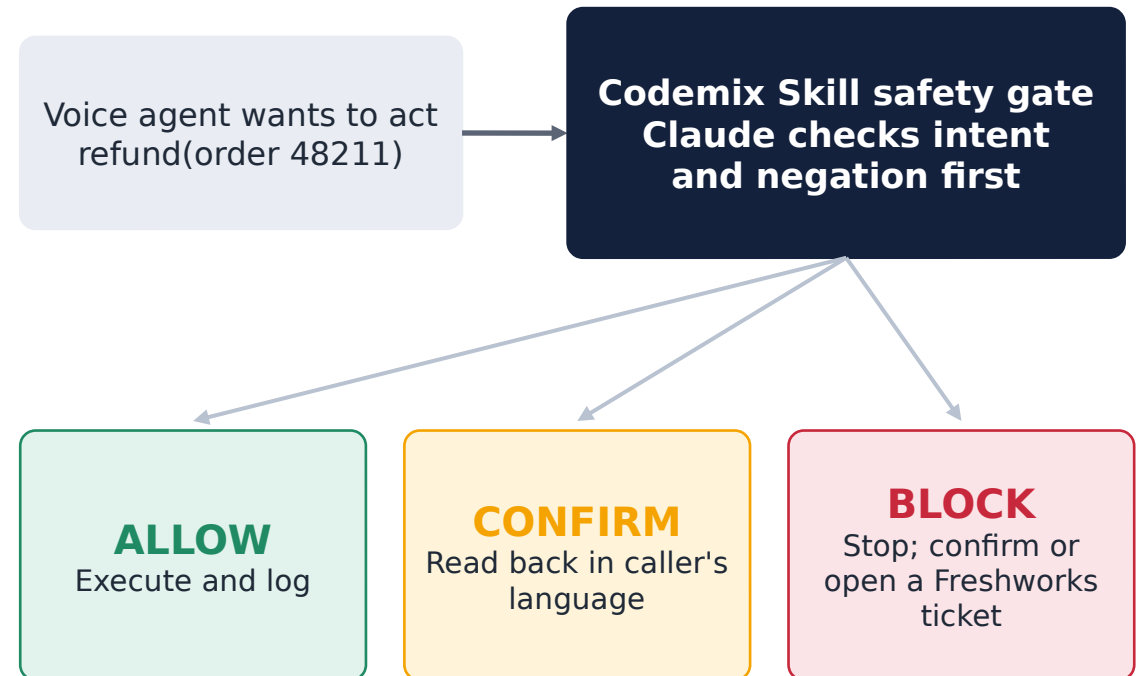
OWNER

Team Ramanathan & Sadhana, builders of Codemix Skill

BUILDS ON

Round-1 code-mix engine + MCP server, live at codemix-skill.vercel.app

How it works



Most actions go straight through. We only ask a question when the action is irreversible and we aren't sure.

Problem Statement

AI support agents now do things, and code-mixed speech is where they will get it wrong

- AI support agents no longer just answer questions. They cancel orders, issue refunds and change addresses on their own.
- Indian callers mix languages mid-sentence, and the meaning often hangs on one small negation word: *mat*, *nahi*, *vendam*, *vaddu*.
- On phone audio, that short word is exactly what speech-to-text mishears or drops.
- The agent then takes the opposite action. It can't be undone, and it costs refunds, reshipping and customer trust.

HINDI NEGATION

Risk: order cancelled

*Order cancel **mat** karo, bas address change karna hai*

Meaning: don't cancel, just change the address

TAMIL NEGATION

Risk: refund issued, no replacement

*Refund **vendam**, replacement anuppunga*

Meaning: I don't want a refund, send a replacement

SPEECH-TO-TEXT DROP

Risk: confident wrong action

Heard: "Order cancel karo..."

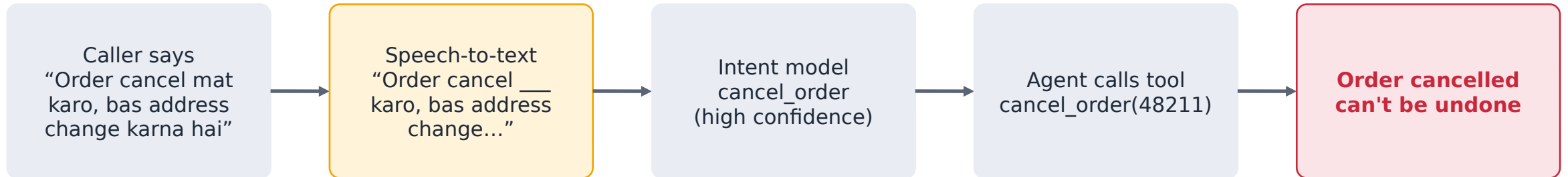
The negation is lost on a noisy line, so the sentence now means the opposite

02

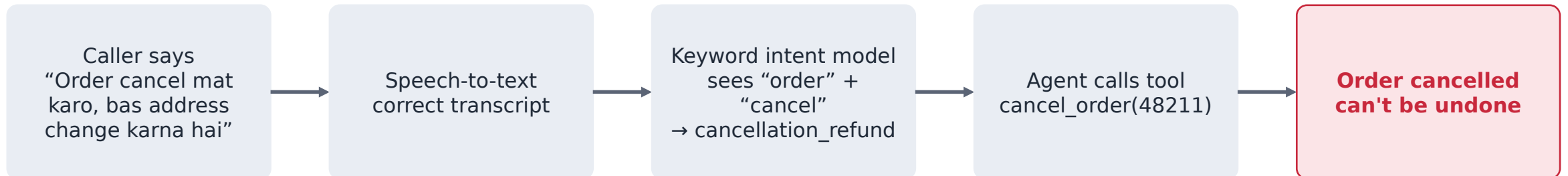
Where It Breaks Today

Two ways the same sentence becomes the wrong action

PATH A: speech-to-text drops the negation



PATH B: transcript is right, but the model ignores "mat"



Measured, not hypothetical: our own Round-1 Codemix Skill engine classifies "Order cancel mat karo, bas address change karna hai" as `cancellation_refund`.

03

Why Now



Agents now have write access

AI agents, tool calling and MCP let a bot cancel, refund or re-address an order directly. A misunderstanding used to produce a bad answer. Now it produces a bad transaction.



Code-mixed voice is normal in India

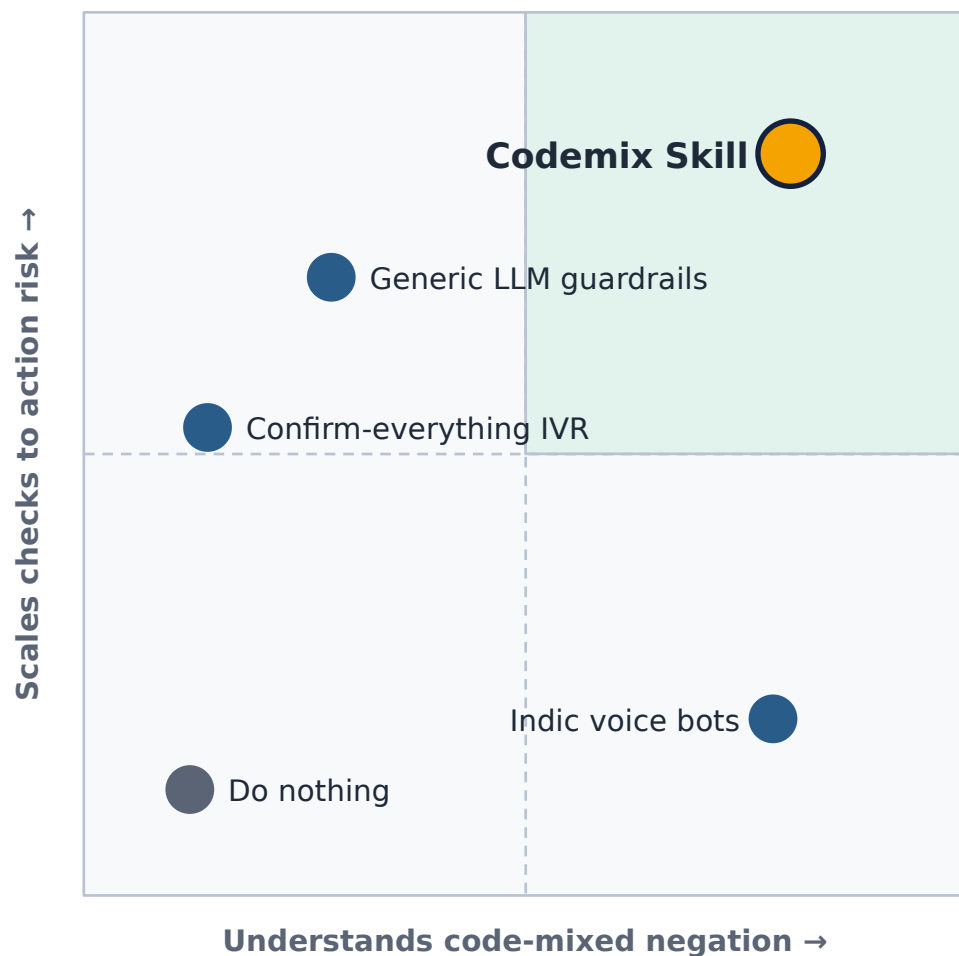
Hinglish and Tanglish callers switch language inside a sentence. Codemix Skill already handles the switching, and negation is the gap we measured in our own engine.



The Indian voice stack is ready

Sarvam AI transcribes code-mixed Indian speech, Claude can reason about intent and negation in real time, and ElevenLabs speaks naturally in the caller's mix. Checking every sensitive action is now affordable.

Competitive Landscape



Alternative

Strength

Gap

Generic LLM guardrails

Good at policy, toxicity and PII filters

Can't tell that "mat" flips "cancel", and ignores speech-to-text confidence

Confirm-everything IVR

Safe: "press 1 to confirm"

Adds friction to every action, so callers drop off

Indic voice-bot platforms

Strong Indian-language speech recognition

Action safety isn't their layer. Tool calls fire unchecked

Do nothing

Zero effort and zero latency

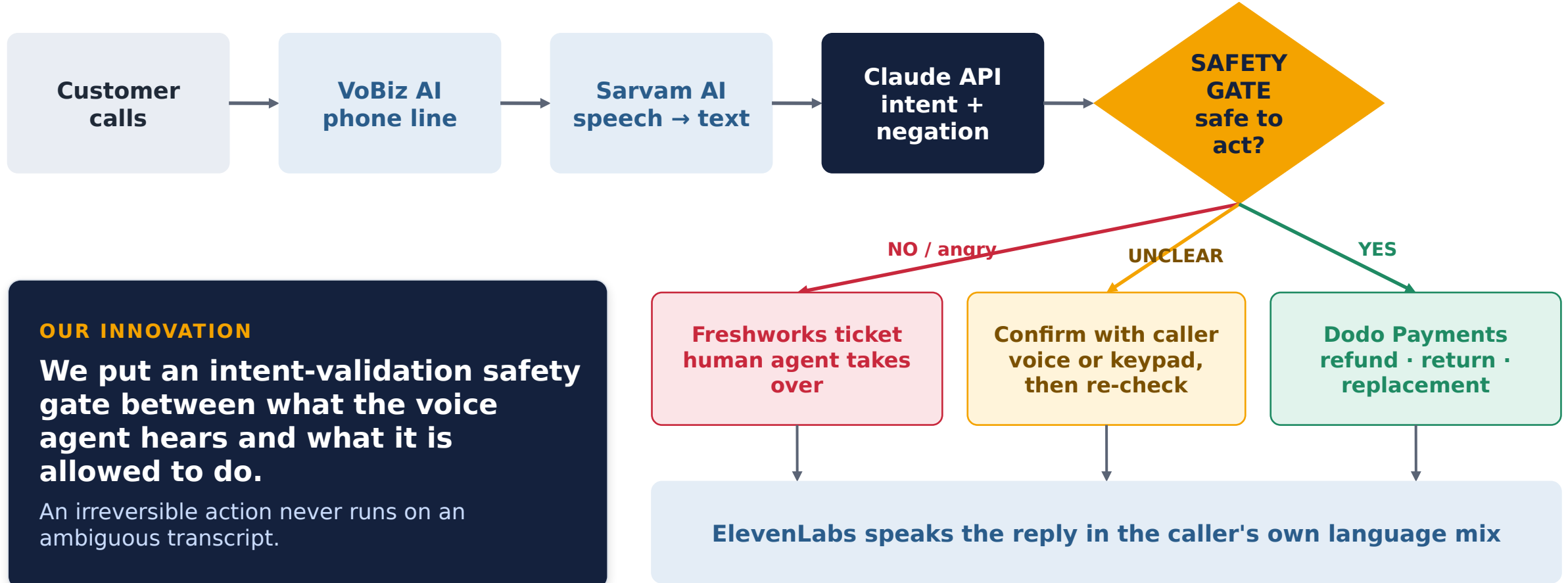
Wrong refunds and cancellations, and customers lose trust

Our edge An intent-validation gate between what the voice agent hears and what it can do, with a Freshworks ticket for anything unsafe.

05

Proposed Solution: the Safety Gate

Every sensitive action passes a safety check between what the voice agent hears and what it does



05 How Each Part Prevents a Wrong Action

Each tool is chosen for the role it plays in stopping a misheard request from becoming a transaction

VoBiz AI

PHONE ACCESS

Gives customers a number to call the AI voice agent, and passes keypad (DTMF) presses through.

Prevents: Callers can confirm with a keypress when the line is noisy.

Sarvam AI

SPEECH-TO-TEXT

Turns spoken code-mixed Hindi + English and Tamil + English into text.

Prevents: Built for Indian speech, so a short “mat” or “vendam” is less likely to be lost.

Claude API

INTENT & SAFETY VALIDATION

Reads the transcript before any sensitive action, finds the real intent and any negation or conflict.

Prevents: This is the safety gate. “Refund vendam” means NO refund.

Dodo Payments

CUSTOMER-SERVICE ACTIONS

Runs refunds, returns and replacements.

Prevents: Only called after the safety gate says YES.

Freshworks

TICKET CREATION

Creates a Freshdesk ticket when the case can't be resolved safely by the bot.

Prevents: Unclear or unsafe cases go to a human, with full context.

ElevenLabs

NATURAL VOICE REPLY

Speaks the answer back in the caller's own language mix.

Prevents: Reads back what it understood, so the caller can catch a mistake.

05

Freshworks Integration

Freshdesk is where every unsafe or unclear case lands, with everything a human agent needs

Safety gate says NO, the caller is angry, or confirmation fails



**Codemix Skill calls the Freshdesk REST API
POST /api/v2/tickets**



**Ticket opens in Freshdesk with an English
summary and the full transcript**



**Human agent resolves it. The bot never guesses
on a risky action**

Freshdesk ticket #1042 · created by Codemix Skill

Subject	Customer declines refund, requests replacement (order 48211)
Priority · Source	High (from P1-P3) · Phone · Status: Open
Tags	codemix-skill · tamil · english
Summary (English)	Caller said “Refund vendam, replacement anuppunga”: no refund, send a replacement
Safety-gate verdict	BLOCK refund · confidence on “vendam” 0.58 · caller didn't confirm NEW
Original transcript	Full code-mixed transcript, both sides, timestamps

Live today: api/create-ticket.js already creates real Freshdesk tickets. **Next:** a Freshworks Marketplace app for any Freshdesk customer.

05

What Codemix Skill Does

Eleven features in four groups



UNDERSTAND

Mixed languages

Sarvam AI + our engine handle Hinglish and Tanglish, even mid-sentence switches

Intent detection

Claude works out what the caller really wants and maps it to the right action

Negation handling

Catches mat, nahi, vendam, vaddu that flip the meaning



DECIDE

Decision layer

ALLOW, CONFIRM or BLOCK for every API action

Transparent uncertainty

Tells the caller what it's unsure of and confirms before acting

Sentiment → human

Angry or distressed callers get a Freshworks ticket and a person



CONVERSE

Low latency

Fast offline checks first; Claude runs only before sensitive actions

DTMF & IVR

VoBiz passes keypad presses: "press 1 to confirm" on any phone

Interrupt handling

The caller can cut in while the agent speaks, and the agent listens



AUDIT

SOP-based auditing

Checks every call against the company's own SOP rules

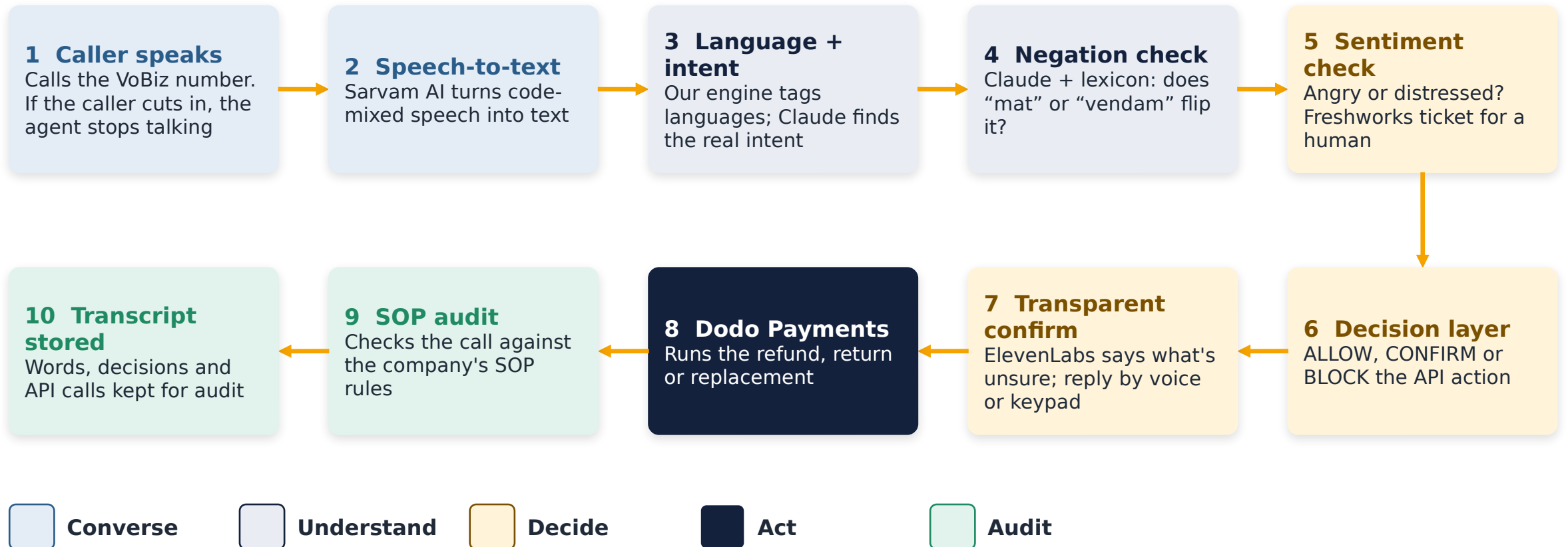
Full audit trail

Stores the transcript, each decision and each API call

05

One Call, End to End

Where each feature acts during a single call

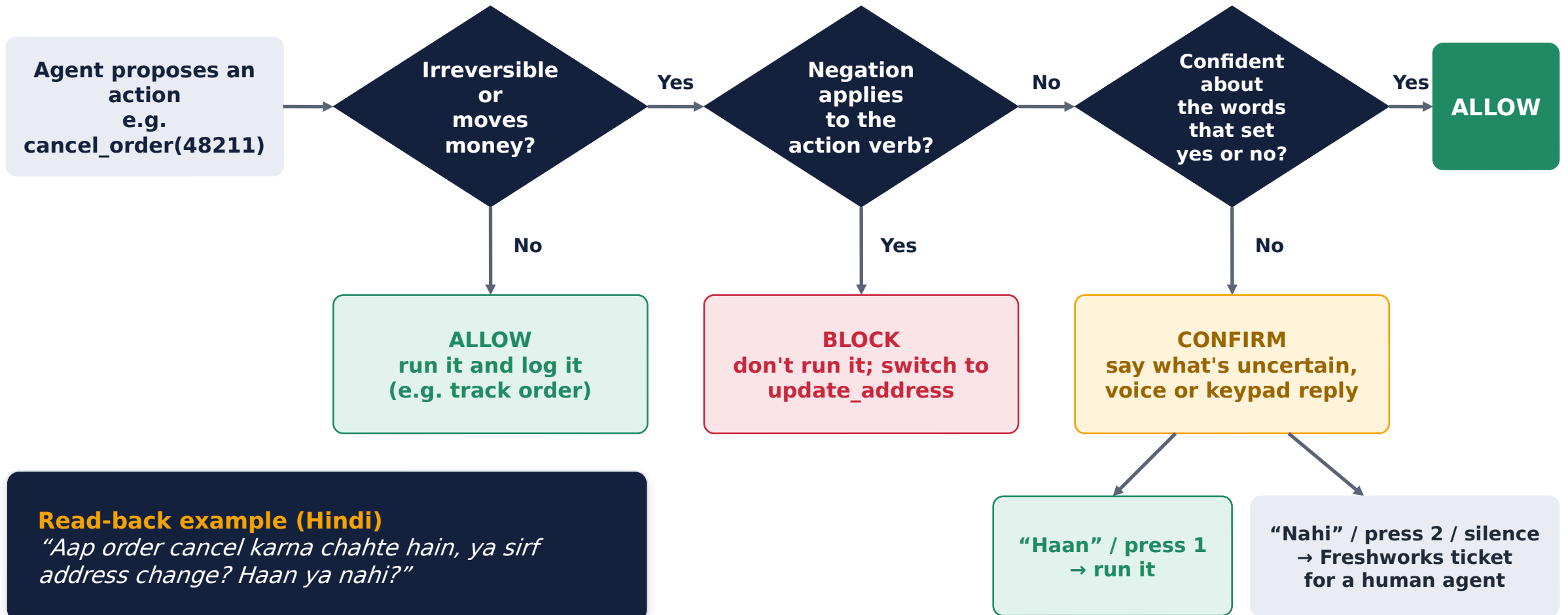


Most calls go 1 → 2 → 3 → 4 → 6 → 8 with no question asked. Steps 5 and 7 only kick in when needed.

05

How the Gate Decides

Only the risky, uncertain branch adds friction for the caller



05

Confirming Before Acting

The agent says what it's unsure of. The caller answers by voice or keypad, and can interrupt at any time

CALLER

"Refund vendam, replacement anuppunga."

AGENT · ElevenLabs voice, in Tamil

"You don't want a refund, you want a replacement for order 48211. I'm not fully sure I heard 'vendam' on this line. Say 'aamaa' or press 1 to confirm. Press 2 to talk to a person."

CALLER · keypad via VoBiz

presses 1

AGENT

"Replacement booked. Reference R-2291."



Transparent uncertainty

Names the exact word or detail it isn't sure of, and how sure it is. No silent guessing on actions that can't be undone.



DTMF & IVR fallback

The keypad works when speech is noisy. VoBiz passes the key press to the agent: press 1 to confirm, 2 for a person.



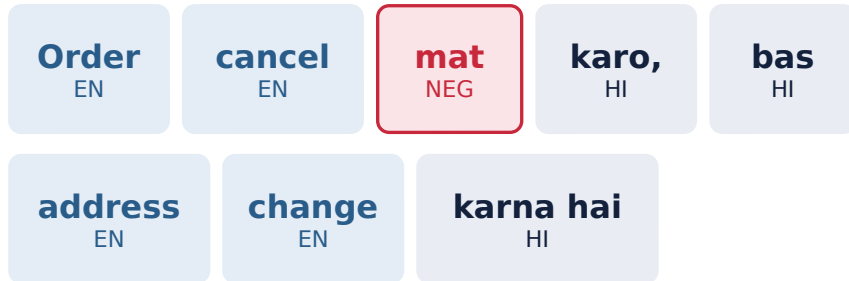
Interrupt handling

If the caller cuts in while the agent is talking, the agent stops speaking and treats the new words as the answer.

05

Two Calls Traced Through the Gate

HINDI CALL



Negation “mat” applies to “cancel karo”

Proposed action cancel_order: irreversible

BLOCK cancel_order → run update_address instead (reversible → ALLOW)

TAMIL CALL



Negation “vendam” applies to “refund”. Speech-to-text confidence on it is only 0.58

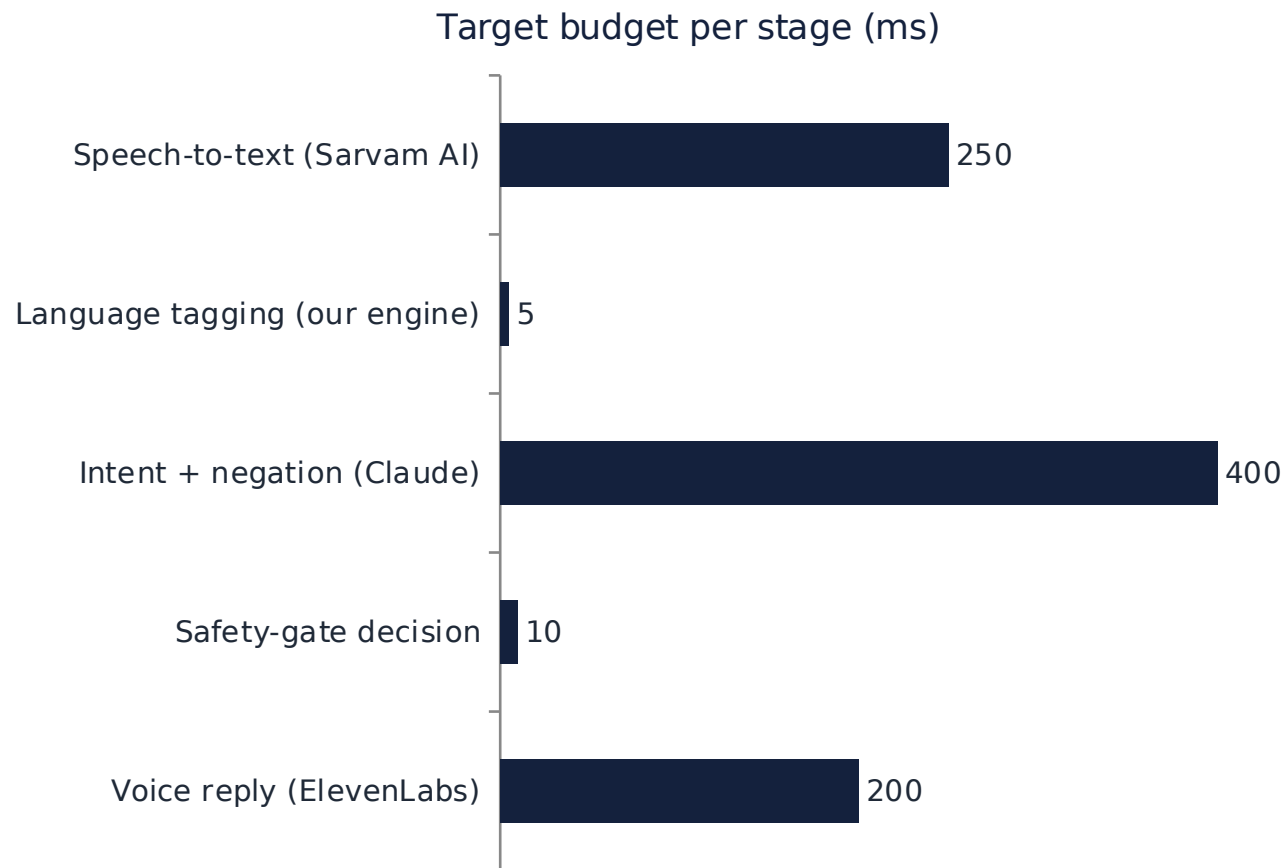
Proposed action issue_refund: moves money

BLOCK the Dodo refund. **CONFIRM** the replacement: “Replacement anuppalaamaa?”

05

Low Latency by Design

Target time budget from the caller's last word to the agent's reply (proposed targets, in ms)



~470 ms

routine replies: no money-moving action,
Claude check skipped

< 900 ms

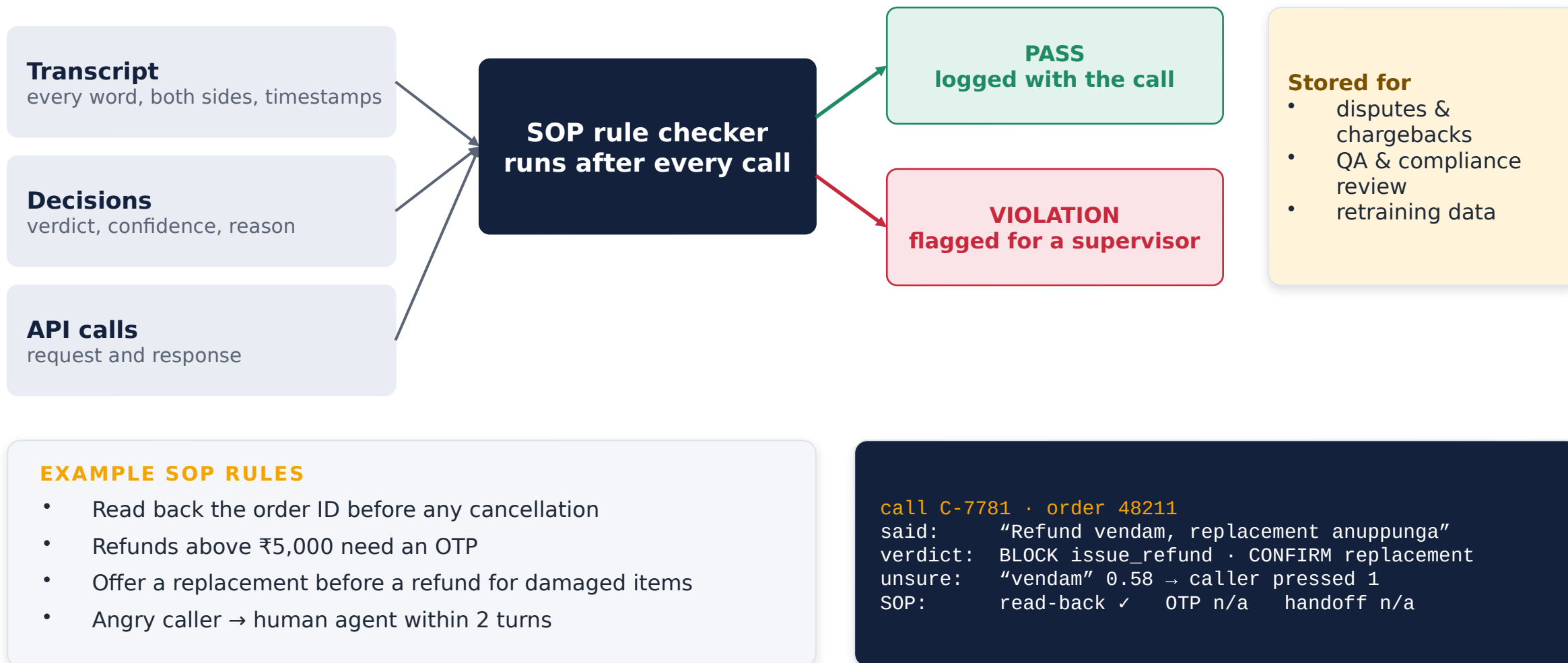
with Claude's safety check before a
refund or cancellation

< 5 ms

our engine's language tagging, in-process

SOP Auditing & Full Audit Trail

Every call is checked against the company's SOP, and the complete record is stored



Scope: MVP vs Later Phases

PHASE 1 · MVP

Months 0-3

- VoBiz + Sarvam AI + ElevenLabs voice
- Claude gate before Dodo Payments
- Freshworks tickets for unsafe cases
- Hindi, Tamil, English + audit

PHASE 2

Months 4-6

- Sentiment-based handoff to human agents
- Interrupt handling tuned for phone lines
- SOP rule editor + audit reports
- Adds Telugu, Kannada, Bengali, Marathi

PHASE 3

Months 7-12

- One-click Freshworks Marketplace app
- Plug-in for any AI agent (MCP / REST)
- Chat and WhatsApp channels



AI & agentic design

Claude acts as the safety agent: it validates intent and negation before any sensitive action. The voice agent can only reach Dodo Payments through the gate, and anything unsafe becomes a Freshworks ticket.



Already built (Round 1)

Code-mix tagging engine, intent scoring, MCP server, a live Freshdesk ticket API, and a demo on Vercel. Round 2 adds the voice path, Claude gate and Dodo actions.

06

Target Customers



PRIMARY

D2C, e-commerce & fintech brands

Indian brands with 20 to 500 support seats, on Freshdesk, starting to let AI voice agents refund, return and replace orders for Hindi- and Tamil-speaking callers.



SECONDARY

BPOs and support outsourcers

They run Freshworks for several Indian brands, and one wrong refund costs them a client. They want one policy layer across all their accounts.

Actions the gate protects



Refund

Moves money through Dodo Payments; costly to reverse



Return / cancellation

Stops an order the caller may still want



Replacement

A wrong one ships an item twice

Value Proposition

Plug it in once, and your AI agent can act for mixed-language callers: it gets the intent right, confirms when unsure, hands angry callers to a person and audits every action.

✗ Without Codemix Skill

- Agent acts on its first guess at the intent
- One dropped “mat” leads to a cancelled order
- Angry callers stay stuck talking to a bot
- No transcript or SOP check to prove what happened

✓ With Codemix Skill

- Understands mixed-language callers and runs the right API
- Checks negation and confidence before any irreversible call
- Says what it's unsure of; caller confirms by voice or keypad
- Angry callers reach a person; every call is SOP-audited

08

Pricing & Packaging

Proposed. We only charge for the actions we actually guard.

Included

Free

for every Freshdesk customer

- Monitor mode: flags risky actions, never blocks
- Hindi + English
- Weekly wrong-action report

Codemix Skill Add-on

\$5

per 1,000 guarded actions

- Claude safety gate, DTMF confirm, interrupts
- Sentiment handoff + SOP audit
- All MVP languages
- First 30 days free

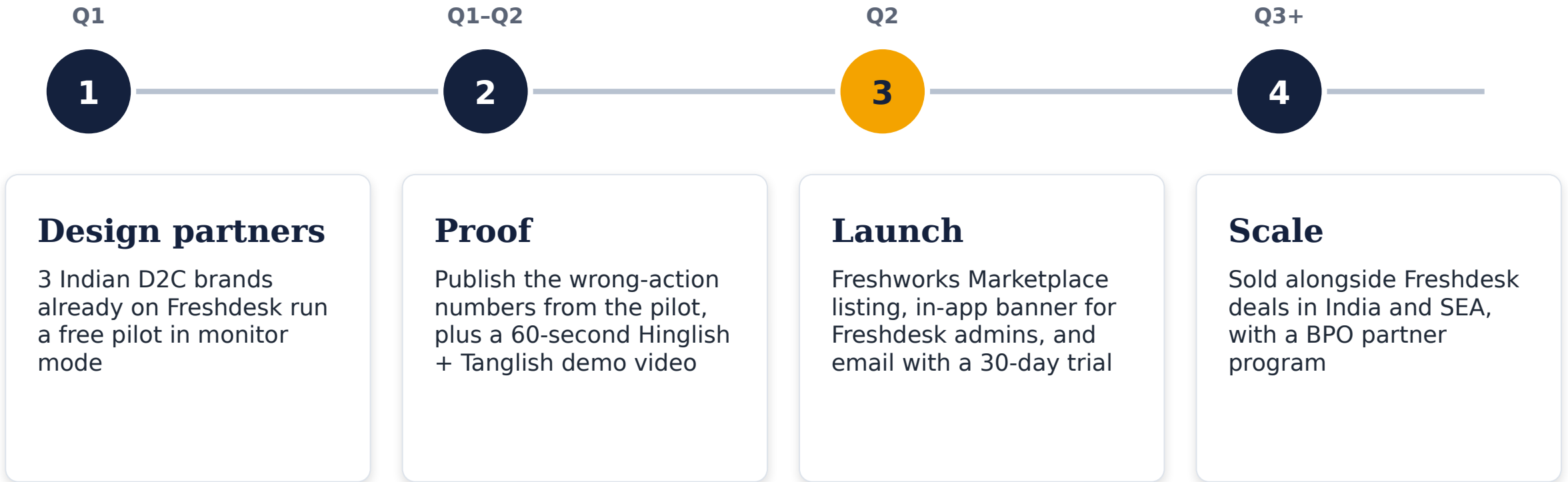
Enterprise

Custom

annual contract

- Custom SOP rule packs
- Transcript retention + data residency in India
- SLA

09 Go-to-Market Plan



Free monitor mode shows each customer their own wrong-action count, and that report is what sells the paid add-on.

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Costs & Resources Needed



3 engineers

voice path, safety gate, Freshworks app



1 linguist / data annotator

Hindi + Tamil negation set



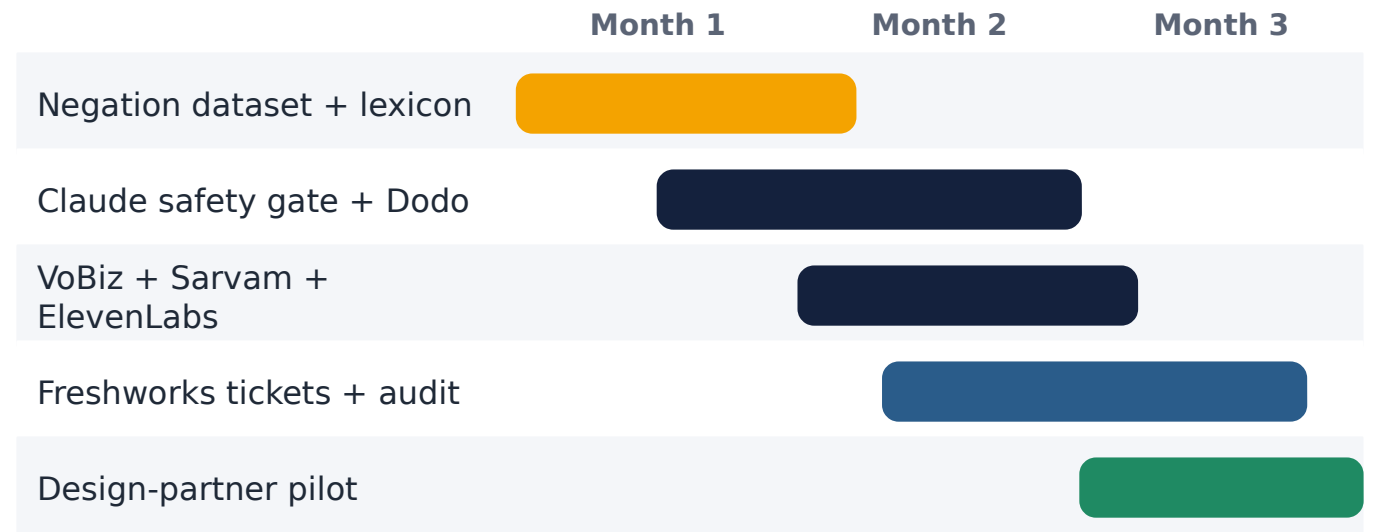
Part-time PM

pilots + pricing

~\$90K

to MVP in 3 months: salaries, Sarvam / Claude / ElevenLabs / VoBiz usage, and labelling ~5,000 negation utterances

MVP timeline



Optional: ARR estimate (assumption-based)

100 paying customers × 20,000 guarded actions / month × \$5 per 1,000 = \$10K MRR, or about **\$120K ARR in Year 1**. Every number here is an assumption to test in the pilot.

11 Expected Benefits & Success Metrics

< 0.5%

Wrong actions

irreversible API calls on negated Hindi + Tamil requests

< 900 ms

Reply time (p95)

from the caller's last word, with the safety check

≤ 15%

Confirmation rate

share of guarded actions that ask a question

100%

Audit coverage

API actions stored with transcript + SOP result

–30%

Reversal tickets

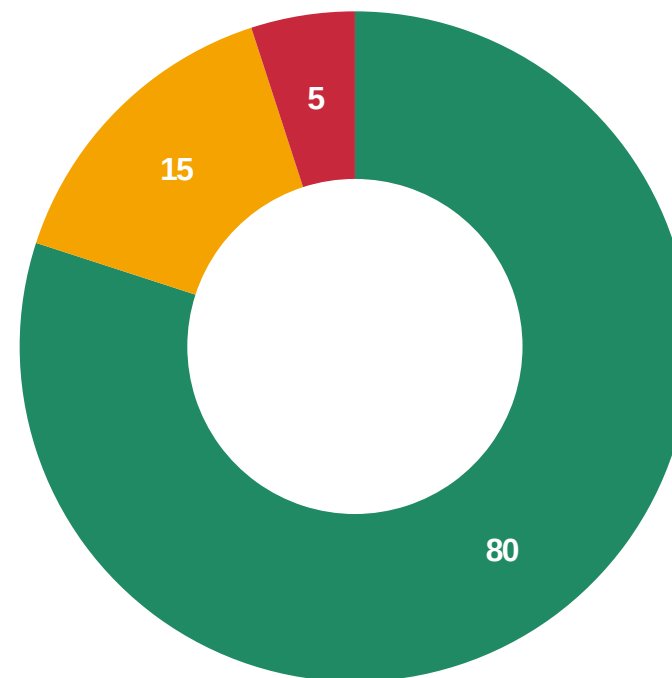
refund / cancellation reversals at pilot customers

15%

Adoption

of eligible voice customers within 2 quarters

Target mix of guarded actions (%)



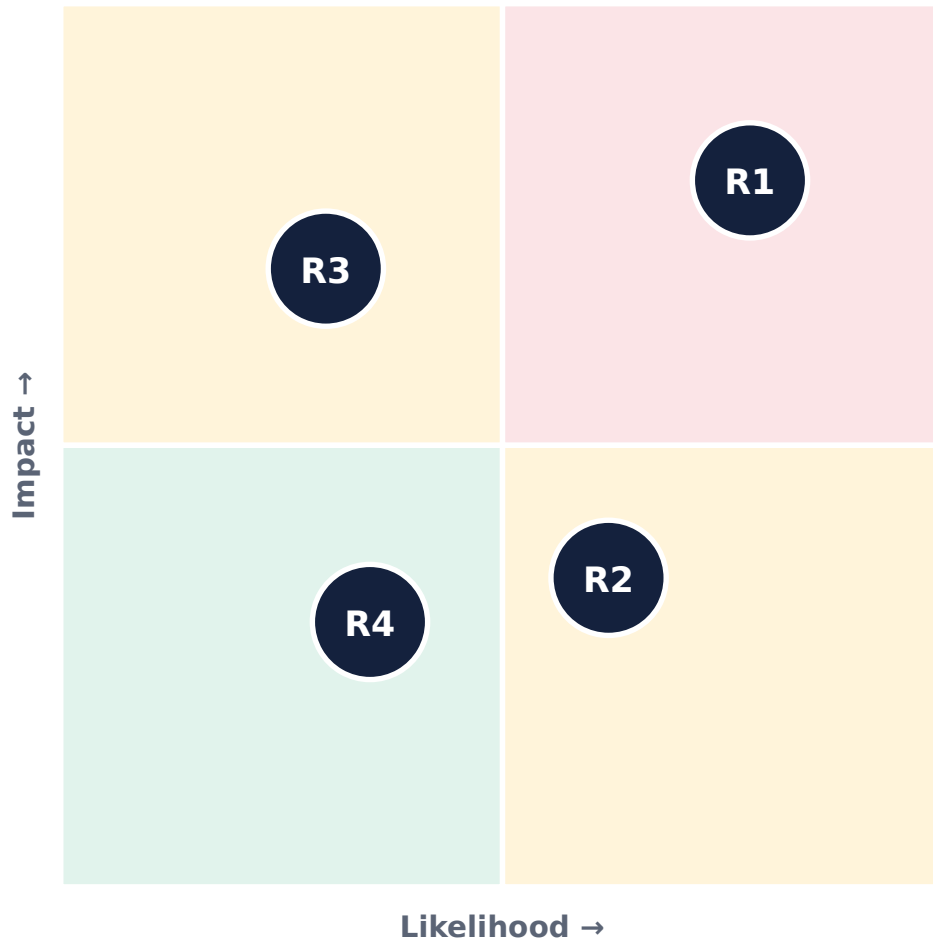
■ ALLOW

■ CONFIRM

■ BLOCK / re-route

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Risks & Assumptions



R1 · Speech-to-text drops the negation completely

Mitigation: for irreversible actions, low confidence on a short word, or a missing word where one is expected, sends the call to CONFIRM

R2 · The safety check adds latency or friction

Mitigation: Claude runs only before sensitive actions; everything else uses the < 5 ms offline path

R3 · Dialects and spellings (vendaam, venaam, matt)

Mitigation: lexicon plus LLM fallback, and every human override is added to the training set

R4 · Assumption: callers accept a read-back

In their own language. We validate with pilot CSAT, and an English read-back is the fallback



“Don't cancel” should mean don't cancel.

We put an intent-validation safety gate between what the voice agent hears and what it is allowed to do.

OUR ASK

A Freshdesk sandbox and a Marketplace listing path for a 3-brand pilot

LIVE TODAY

codemix-skill.vercel.app: the Round-1 engine this builds on